Google Cloud H2S

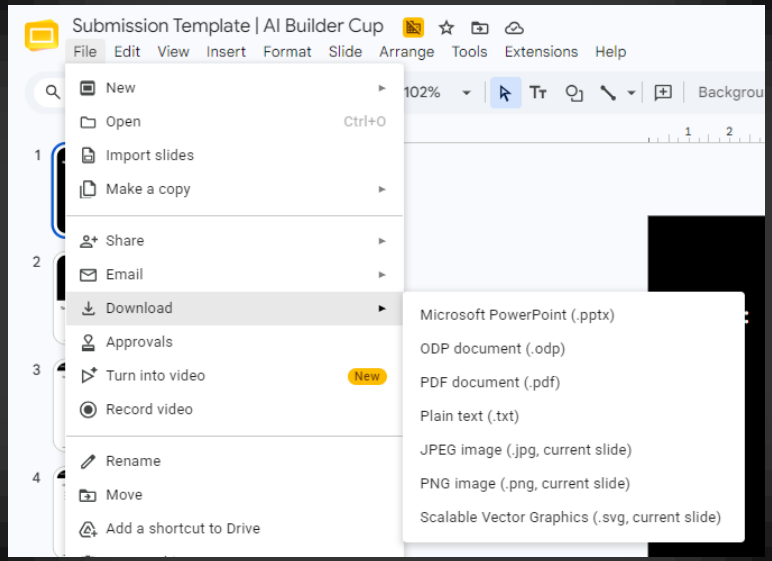
AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Neural Nuggets**
b. Team leader name: **Kyaw Kyaw Oo**
c. Problem Statement: **Media, Content & Digital Experiences:** Personalized learning by teaching an AI student



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

Brief about the idea

Teach Milo

A web app where learners teach a curious AI student using their own study material.

Who it serves

Students and self-learners who want to practise explaining and applying a concept.

How it works

Reading can make a page feel familiar while the idea is still hard to explain, so gaps stay hidden. Explaining an idea to someone else shows what you really know. Milo asks about gaps, presents a misconception to correct, and offers a fresh application task.

Working prototype on Google Cloud with Gemini. Guests can start without an account.

Opportunities

How it differs

The AI takes the student role. The learner explains material from their own notes instead of receiving a generated lesson to read.

How it addresses the problem

Questions and deliberate misconceptions expose gaps. An independent check asks the learner to apply the idea in a new situation.

USP

Teaching, an independent application task, and feedback that quotes the learner’s own explanation in one session.

List of features offered by the solution

Functional features

Your own material

Upload PDF, DOCX, PPTX or TXT, or paste notes.

Review and prepare

Correct extracted text. Choose focus, depth and language.

Teach in chat and on boards

Explain ideas and correct Milo’s misconception.

Independent check and report

Try a fresh task and inspect cited feedback.

Non-functional features

Validated AI output

Schemas check replies. The server verifies calculations.

Protected data and secrets

Server-side keys, hashed passwords, 7-day data expiry.

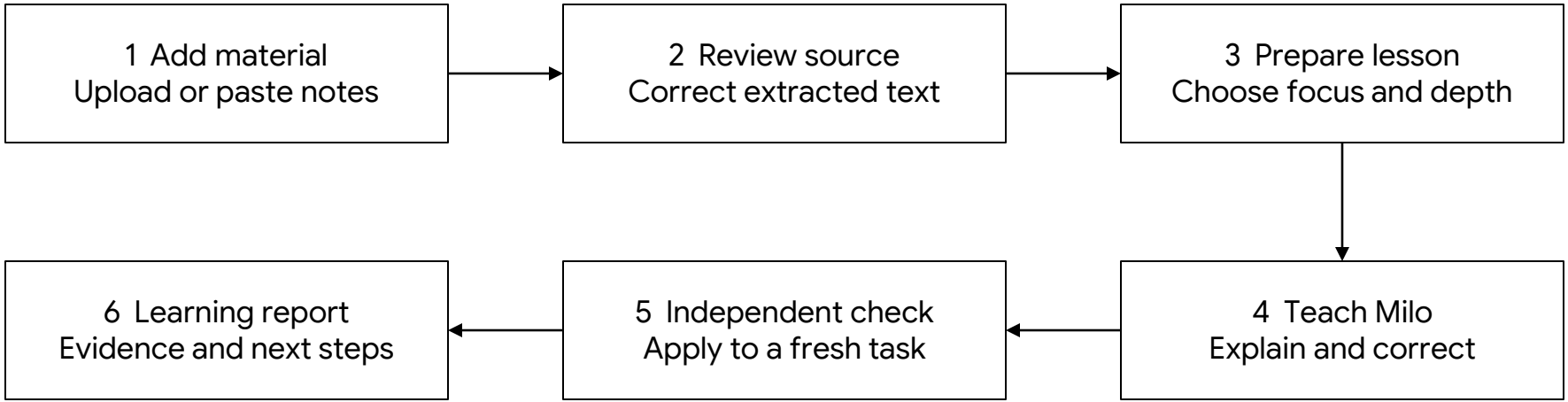
Reusable teaching boards

Activities, concept maps, matching, tables and sketches.

Language and appearance

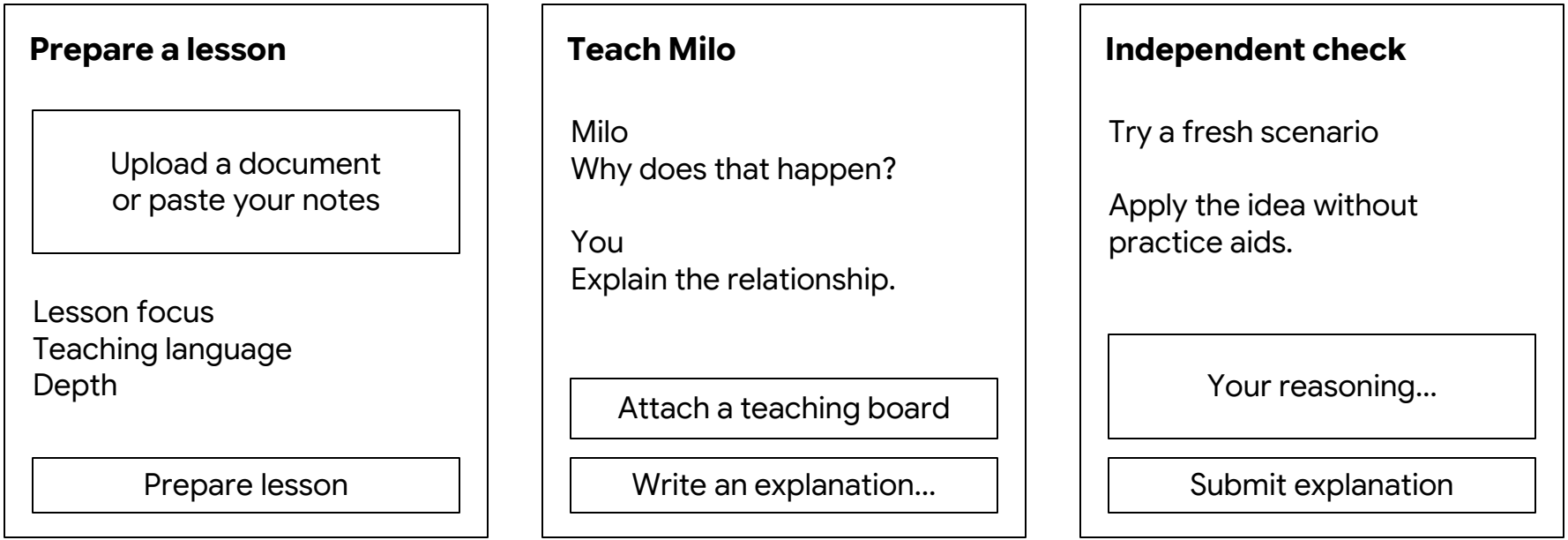
5 interface languages, 9 teaching languages, light/dark.

Process flow diagram or Use-case diagram



Teaching can continue until the evidence supports readiness. The independent check is optional. Learners can end a session at any time and receive feedback.

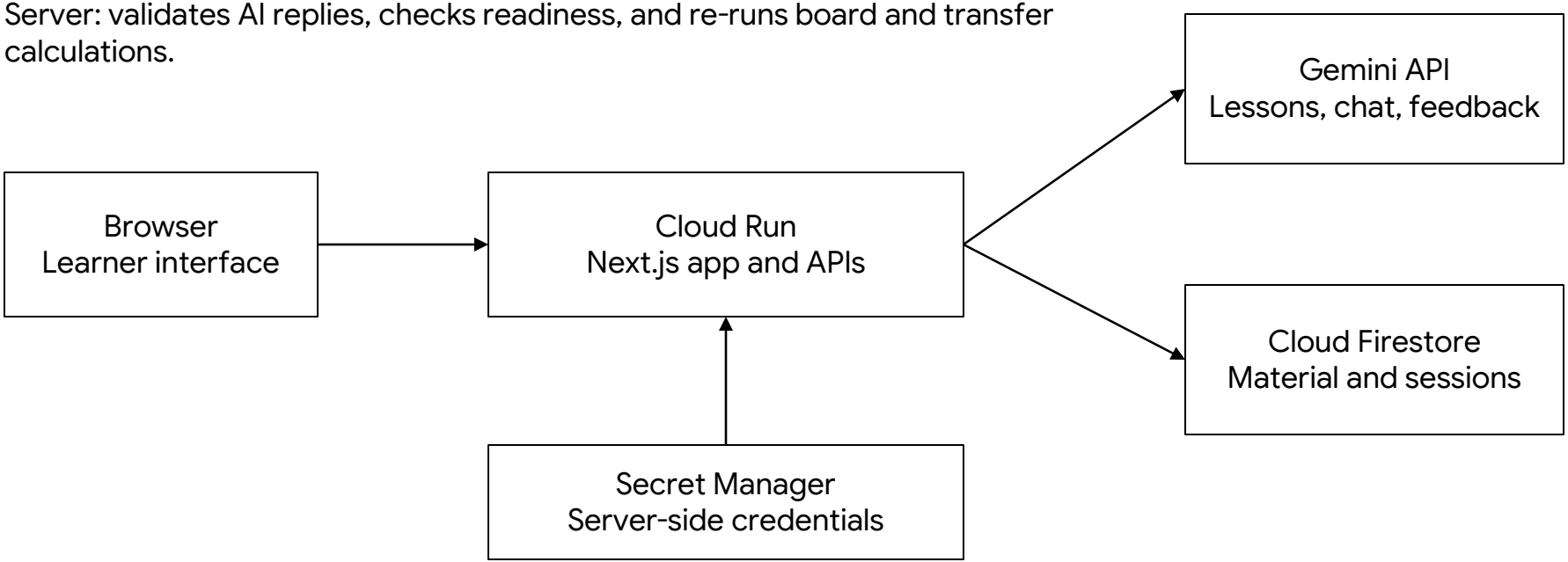
Wireframes/Mock diagrams of the proposed solution (optional)



Schematic layouts. Actual prototype screenshots appear on slide 11.

Architecture diagram of the proposed solution

Server: validates AI replies, checks readiness, and re-runs board and transfer calculations.



Cloud Build and Artifact Registry deploy the app to Cloud Run in Singapore.

Technologies to be used in the solution

Technology	Role in the prototype
Gemini and Google Gen AI SDK	Extract content and generate lessons, activities, chat and feedback
Next.js, React and TypeScript	Website and server API routes
Cloud Run	Host the containerized app in Singapore
Cloud Firestore	Store material, lessons, conversations and evidence with expiry
Cloud Build and Artifact Registry	Build and deploy the application container
Secret Manager	Keep API keys and authentication secrets on the server
Zod and calculation sandbox	Validate structured replies and verify numeric calculations

Estimated implementation cost (optional)

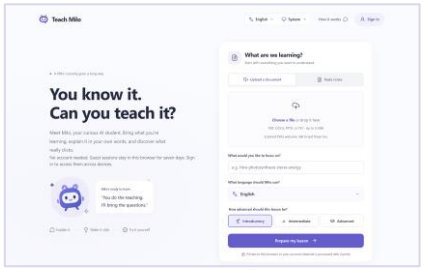
Free-tier inference: US$0 only for eligible use within the project's actual quota.

Illustrative paid equivalent for Gemini 3.5 Flash-Lite, standard rates

Assumption / calculation	1,000 illustrative sessions
12 calls × 1,500 input tokens per session	18M input tokens × US$0.30/M = US$5.40
12 calls × 450 output tokens per session	5.4M output tokens × US$2.50/M = US$13.50
Gemini API only	US$18.90 total, or US$0.0189 per session

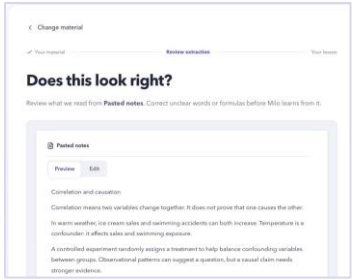
Usage assumptions are illustrative. Output includes thinking tokens. Hosting, Firestore and deployment charges are separate. The benchmark on slide 12 reports observed API usage.

Snapshots of the prototype



Add your material

Upload a file or paste notes



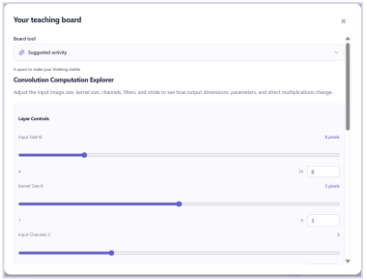
Check the extracted text

Fix words or formulas first



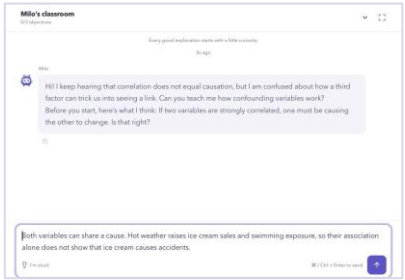
Preview Milo's lesson

Topic, depth and goals



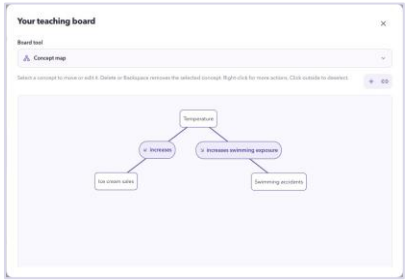
Suggested activity board

Built from your material



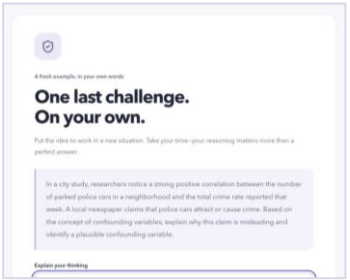
Teach Milo

Correct Milo's mistaken idea



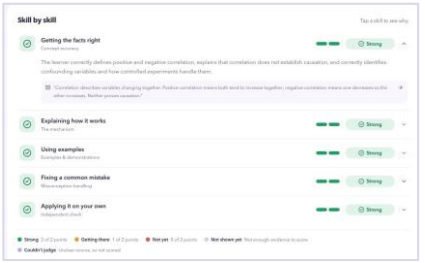
Teaching boards

Concept map, matching, sketch



Independent check

Apply it to a fresh task



Learning report

Feedback quotes your words

Prototype Performance report/Benchmarking

gemini-3.5-flash-lite. Direct Gemini SDK calls, n=9, concurrency 1.

Workload	API OK	p50 / max latency	Input / output tokens (total)
lesson preparation	3/3	5.20 s / 5.80 s	5,793 / 5,008
teaching reply	3/3	2.50 s / 2.70 s	3,513 / 1,624
report evaluation	3/3	4.07 s / 4.21 s	3,849 / 3,918

Structured JSON: 9/9. Fixture checks: 9/9. Paid equivalent: US$0.02991 total.

Paced throughput: 2.76 successful calls/min with 20 s gaps. No capacity claim.

Constraints: active RPM/TPM/RPD and account tier unverified. No quota increase. Free-tier charge is US$0 only if eligible. This run excludes app/HTTP latency, PDF OCR and concurrent load. Fixture checks do not establish general AI accuracy.

18 Oct 2026. macOS arm64, Node 26.11.1. Raw logs in project.

Additional Details/Future Development (if any)

Available today

Gemini builds activities from a catalog of 17 reusable components. Learners can attach a board to their written explanation.

Next: measure the complete learner journey

Instrument app response times, PDF OCR, retries and tokens per completed session. Repeat across document sizes and teaching languages.

Next: validate learning and expand practice

Run a learner pilot with independent application tasks. Add focused follow-up activities and recurring practice using the shared activity catalog.

Provide links to your:

GitHub Public Repository

https://github.com/kyawoo-23/teach-milo

Demo Video Link (3 Minutes)

https://youtu.be/ZchsNx0_S80

Final Product Link

https://teach-milo-86758133633.asia-southeast1.run.app

Google Cloud H2S

AI Builder Cup

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Build for the world.

Thank you

